



C23-M-MRAC-302

23143

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DME – THIRD SEMESTER EXAMINATION

APPLIED ELECTRICAL AND ELECTRONICS

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **three** marks.
(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. State the laws of resistance.
2. State the advantages of three-phase system over single-phase system.
3. List the different types of D.C. motors.
4. List the applications of single-phase induction motors.
5. List the types of servo motor.
6. Define sensor and list its types.
7. State the need of fuse.
8. State different preventive methods employed to avoid electric shock.
9. Define the terms (a) conductor (ii) semi-conductor and (c) insulator.
10. List different transistor configurations.

PART—B

10×5=50

Instructions : (1) Answer *any five* questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

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| 11. | State the explain Kirchhoff's Laws. | 10 |
| 12. | Explain the working of transformer with neat sketch. | 10 |
| 13. | Explain the working 3-phase Induction motor with neat sketch. | 10 |
| 14. | Explain the working principle of DC Servo motor with neat sketch. | |
| 15. | Define PLC and write its advantages and its applications. | |
| 16. | Explain the procedure of plate earthing with neat sketch. | 10 |
| 17. | (a) Distinguish between Intrinsic and Extrinsic Semiconductors. | 6 |
| | (b) List the applications of transistors. | 4 |
| 18. | Explain the working of Zener diode and draw its V-I characteristics. | 10 |

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